

DSA0503 - Query Processing for Data Science

Activity

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Activity using SQLite Database:

1. Create the Department Table (Parent Table) with attributes Department_ID INTEGER PRIMARY KEY, Department_Name TEXT
2. Create the Student Table (Child Table) with the following attributes:
Student_ID INTEGER PRIMARY KEY,
Name TEXT,
Age INTEGER,
Department_ID INTEGER,
CGPA REAL,
FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)
3. Display students from CSE department.
4. Count students in each department.
5. Update the foreign key. For eg., change the department of a student
6. Delete a department
7. Display the student's name, student's department and CGPA if the student's CGPA is greater than 8.

1. Create the Department Table (Parent Table) with attributes Department_ID

INTEGER PRIMARY KEY, Department_Name TEXT

```
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Generate + Code + Markdown

cursor.execute("create table department(department_id integer primary key, department_name text)")
[4] ✓ 0.0s Python
... <sqlite3.Cursor at 0x1886d05b340>
```

2. Create the Student Table (Child Table) with the following attributes:

Student_ID INTEGER PRIMARY KEY, Name TEXT, Age INTEGER, Department_ID
INTEGER, CGPA REAL, FOREIGN KEY (Department_ID) REFERENCES
Department(Department_ID)

```
2. Create the Student Table (Child Table) with the following attributes: Student_ID INTEGER PRIMARY KEY, Name TEXT, Age INTEGER,
Department_ID INTEGER, CGPA REAL, FOREIGN KEY (Department_ID) REFERENCES Department(Department_ID)

cursor.execute("create table student(student_id integer primary key, student name text, department_id integer , CGPA real , "
" foreign key(department_id) references department(department_id))")
✓ 0.0s Python
<sqlite3.Cursor at 0x1886d05b340>
```

3. Display students from CSE department.

```
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cursor.execute("Select * from student Join department on student.department_id = department.department_id where
department.department_name = 'CSE'");
row = cursor.fetchall()
for i in row:
    print(i)
[14] ✓ 0.0s Python
... (101, 'Alice', 1, 8.9, 1, 'CSE')
(102, 'Bob', 1, 8.4, 1, 'CSE')
```

4.Count students in each department.

```
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cursor.execute("Select department.department_name,count(student.student_id) from department,student where department.
department_id = student.department_id group by department.department_name");
row = cursor.fetchall()
for i in row:
    print(i)
```

[15] ✓ 0.0s Python

```
... ('CSE', 2)
    ('ECE', 1)
    ('EEE', 1)
    ('IT', 1)
```

5. Update the foreign key. For eg., change the department of a student

```
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cursor.execute("""UPDATE Student SET Department_ID = 2 WHERE Student_ID = 101 """)
```

[16] ✓ 0.0s Python

```
... <sqlite3.Cursor at 0x1886d05b340>
```

```
cursor.execute("Select * from student")
row = cursor.fetchall()
for i in row:
    print(i)
```

[18] ✓ 0.0s Python

```
... (101, 'Alice', 2, 8.9)
    (102, 'Bob', 1, 8.4)
    (103, 'Charlie', 2, 9.1)
    (104, 'David', 3, 7.8)
    (105, 'Eva', 5, 9.3)
```

6. Delete a department

```
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cursor.execute("""DELETE FROM Department WHERE Department_ID = 3 """)
```

[19] ✗ 0.0s Python

```
... -----
IntegrityError                                Traceback (most recent call last)
Cell In[19], line 1
----> 1 cursor.execute("""DELETE FROM Department WHERE Department_ID = 3 """)

IntegrityError: FOREIGN KEY constraint failed
```

A foreign key is used to maintain the relationship between two tables. You cannot delete a parent record if it is still being referenced by a child table, because it would leave invalid or orphaned records.

7. Display the student's name, student's department and CGPA if the student's CGPA is greater than 8.

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```

```
cursor.execute("Select * from student join department on student.department_id = department.department_id where student.CGPA > 8")
row = cursor.fetchall()
for i in row:
    print(i)
```

[24] ✓ 0.0s Python

```
... (101, 'Alice', 2, 8.9, 2, 'ECE')
      (102, 'Bob', 1, 8.4, 1, 'CSE')
      (103, 'Charlie', 2, 9.1, 2, 'ECE')
      (105, 'Eva', 5, 9.3, 5, 'IT')
```